

一、填空题 (共 5 小题, 每小题 4 分, 共计 20 分)

(1) 2. (2) dx. (3) $\sqrt{2}(e^{\pi}-1)$. (4) $\frac{\pi}{2}$. (5) $xy = e^x + C$.

二、选择题 (共 5 小题, 每小题 4 分, 共计 20 分)

(6) D. (7) D. (8) C. (9) B. (10) A.

三、计算题 (共 5 小题, 每小题 6 分, 共计 30 分)

(11) 求极限 $\lim_{x \rightarrow 0} \frac{x - \tan x}{\tan x - \sin x}$.

解: $\lim_{x \rightarrow 0} \frac{x - \tan x}{\tan x - \sin x} = \lim_{x \rightarrow 0} \frac{x - \tan x}{\tan x(1 - \cos x)} = \lim_{x \rightarrow 0} \frac{x - \tan x}{\frac{x^3}{2}} = \lim_{x \rightarrow 0} \frac{1 - \sec^2 x}{\frac{3}{2}x^2} = -\frac{2}{3}$.

(12) 计算不定积分 $\int \ln^2 x dx$.

解: $\int \ln^2 x dx = x \ln^2 x - 2 \int \ln x dx = x \ln^2 x - 2(x \ln x - \int x dx) + C = x \ln^2 x - 2(x \ln x - x) + C$.

(13) 计算定积分 $\int_0^2 \frac{2x+1}{x^2-2x+2} dx$.

解: $\int_0^2 \frac{2x+1}{x^2-2x+2} dx = \int_0^2 \frac{2x-2+3}{x^2-2x+2} dx = \int_0^2 \left[\frac{(x^2-2x+2)'}{x^2-2x+2} + \frac{3}{(x-1)^2+1} \right] dx$
 $= [\ln(x^2-2x+2) + 3 \arctan(x-1)]_0^2 = \frac{3\pi}{2}$.

另解令 $t = x-1$, 则

$\int_0^2 \frac{2x+1}{x^2-2x+2} dx = \int_{-1}^1 \frac{2t+3}{t^2+1} dt = \int_{-1}^1 \frac{2t}{t^2+1} dt + \int_{-1}^1 \frac{3}{t^2+1} dt = 0 + 6 \arctan t \Big|_{-1}^1 = \frac{3\pi}{2}$.

(14) 求微分方程 $y'' - 2y' - 3y = 1 - 3x$ 的通解.

解: 齐次方程 $y'' - 2y' - 3y = 0$ 的特征方程为 $r^2 - 2r - 3 = 0$, 特征根为 $r_1 = -1, r_2 = 3$,

故齐次方程的通解为 $Y(x) = C_1 e^{-x} + C_2 e^{3x}$.

设非齐次方程的特解为 $y^* = a + bx$, 代入原方程得

$3bx - 2b - 3a = 1 - 3x$, 解得 $a = -1, b = 1$, 即 $y^* = x - 1$,

故通解为 $y(x) = C_1 e^{-x} + C_2 e^{3x} + x - 1$.

(15) 设函数 $y = y(x) = x^3 + ax^2 + bx + c$ 在 $x = 2$ 处取得极值, 且点 $(1, 3)$ 是曲线 $y = y(x)$

的拐点, 求常数 a, b, c 的值.

解: $y'(x) = 3x^2 + 2ax + b$, $y''(x) = 6x + 2a$, 由题意知

$$a + b + c = 2, 12 + 4a + b = 0, 6 + 2a = 0, \text{解得 } a = -3, b = 0, c = 5.$$

四、解答题 (共 2 小题, 每小题 8 分, 共计 16 分)

(16) 求函数 $f(x) = \int_0^{x^2} (2-t)e^{-t} dt$ 的最值.

$$\text{解: } f(x) = \int_0^{x^2} (2-t)e^{-t} dt = \int_0^{x^2} (t-2)de^{-t} = (t-1)e^{-t} \Big|_0^{x^2} = (x^2-1)e^{-x^2} + 1,$$

$$f'(x) = 2x(2-x^2)e^{-x^2}, \text{得驻点 } x=0, x=\sqrt{2}, x=-\sqrt{2}, \text{则 } f(0)=0,$$

$$f(\sqrt{2}) = f(-\sqrt{2}) = e^{-2} + 1,$$

$$\text{又 } \lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} (x^2-1)e^{-x^2} + 1 = 1,$$

$$\text{则最大值为 } f(\sqrt{2}) = f(-\sqrt{2}) = e^{-2} + 1, \text{最小值为 } f(0) = 0.$$

(17) 设曲线方程为 $y = \frac{1}{1+x^2}$, 求① 曲线的渐近线; ② 曲线与其渐近线所夹平面区域的

面积; ③ 曲线绕其渐近线一周所围立体区域的体积.

解: ① 因为 $\lim_{x \rightarrow \infty} \frac{1}{1+x^2} = 0$, 则有水平渐近线 $y=0$, 无铅直和斜渐近线;

$$\text{② 面积 } A = \int_{-\infty}^{+\infty} \frac{1}{1+x^2} dx = \arctan x \Big|_{-\infty}^{+\infty} = \frac{\pi}{2} - \left(-\frac{\pi}{2}\right) = \pi;$$

$$\text{③ 体积 } V = \pi \int_{-\infty}^{+\infty} \frac{1}{(1+x^2)^2} dx = \pi \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\sec^2 t}{(\sec^2 t)^2} dt = 2\pi \int_0^{\frac{\pi}{2}} \cos^2 t dt = 2\pi \cdot \frac{1}{2} \cdot \frac{\pi}{2} = \frac{\pi^2}{2}.$$

五、证明题 (共 2 小题, 每小题 7 分, 共计 14 分)

(18) 判断函数 $f(x) = \frac{\ln x}{x}$ 在 $(0, +\infty)$ 上的单调性, 并由此证明 $e < a < b$ 时, $a^b > b^a$.

$$\text{解: } f'(x) = \frac{1-\ln x}{x^2}, \text{则在 } (0, e) \text{ 上 } f'(x) > 0, \text{函数单调增加;}$$

$$\text{则在 } [e, +\infty) \text{ 上 } f'(x) < 0, \text{函数单调减少.}$$

$$\text{又当 } e < a < b \text{ 时, } f(a) = \frac{\ln a}{a} > f(b) = \frac{\ln b}{b}, \text{即 } b \ln a > a \ln b, \text{从而 } a^b > b^a.$$

(19) (1) 设函数 $f(x)$ 在 $[0, 1]$ 上连续, 证明: $\int_0^\pi x f(\sin x) dx = \frac{\pi}{2} \int_0^\pi f(\sin x) dx$;

(2) 计算定积分 $\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx$.

$$\begin{aligned} \text{解: } \int_0^{\pi} x f(\sin x) dx &= \int_{\pi}^0 (\pi-t) f[\sin(\pi-t)] d(\pi-t) = \pi \int_0^{\pi} f(\sin t) dt - \int_0^{\pi} t f(\sin t) dt \\ &= \pi \int_0^{\pi} f(\sin x) dx - \int_0^{\pi} x f(\sin x) dx, \end{aligned}$$

$$\text{移项并求得 } \int_0^{\pi} x f(\sin x) dx = \frac{\pi}{2} \int_0^{\pi} f(\sin x) dx.$$

$$\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx = \frac{\pi}{2} \int_0^{\pi} \frac{\sin x}{1 + \cos^2 x} dx = -\frac{\pi}{2} \int_0^{\pi} \frac{1}{1 + \cos^2 x} d \cos x = -\frac{\pi}{2} \arctan \cos x \Big|_0^{\pi} = \frac{\pi^2}{4}.$$